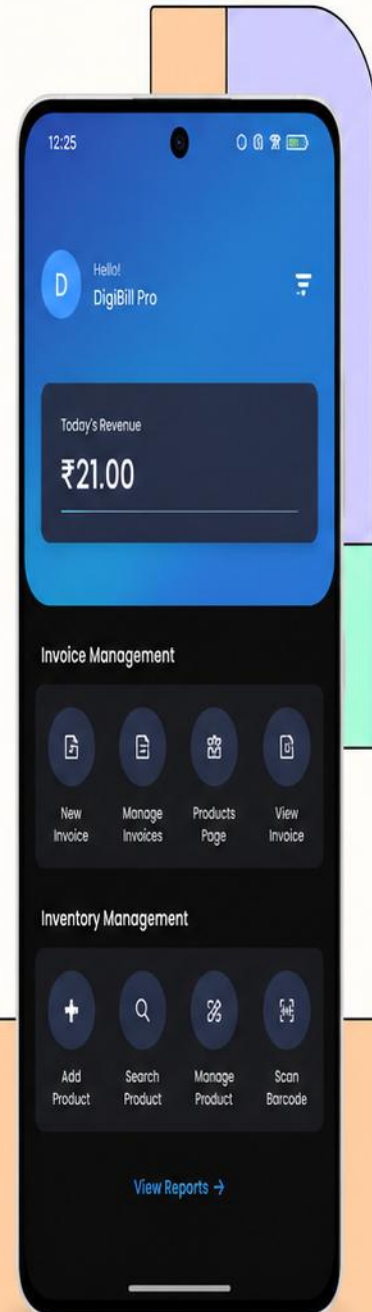




DigiBill

An Invoice App

DigiBill is an affordable and efficient business management solution designed specifically for small business owners who cannot afford or prefer not to invest in expensive hardware.



Introduction

1.

Designing

3.

Future Scope and Planning

5.

**Let's See
what we're
going to
discuss today**

2.

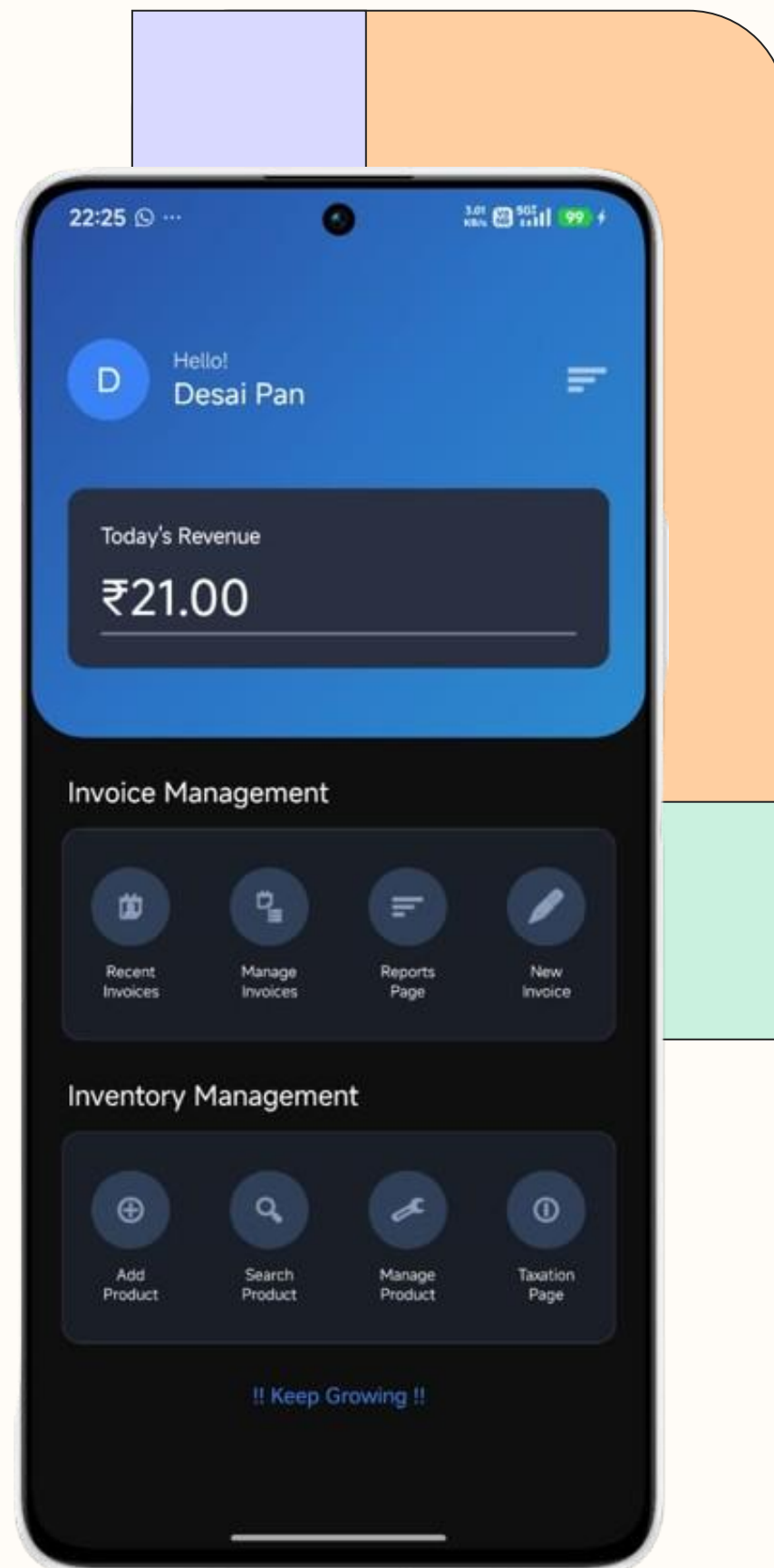
Project Overview

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Conclusion

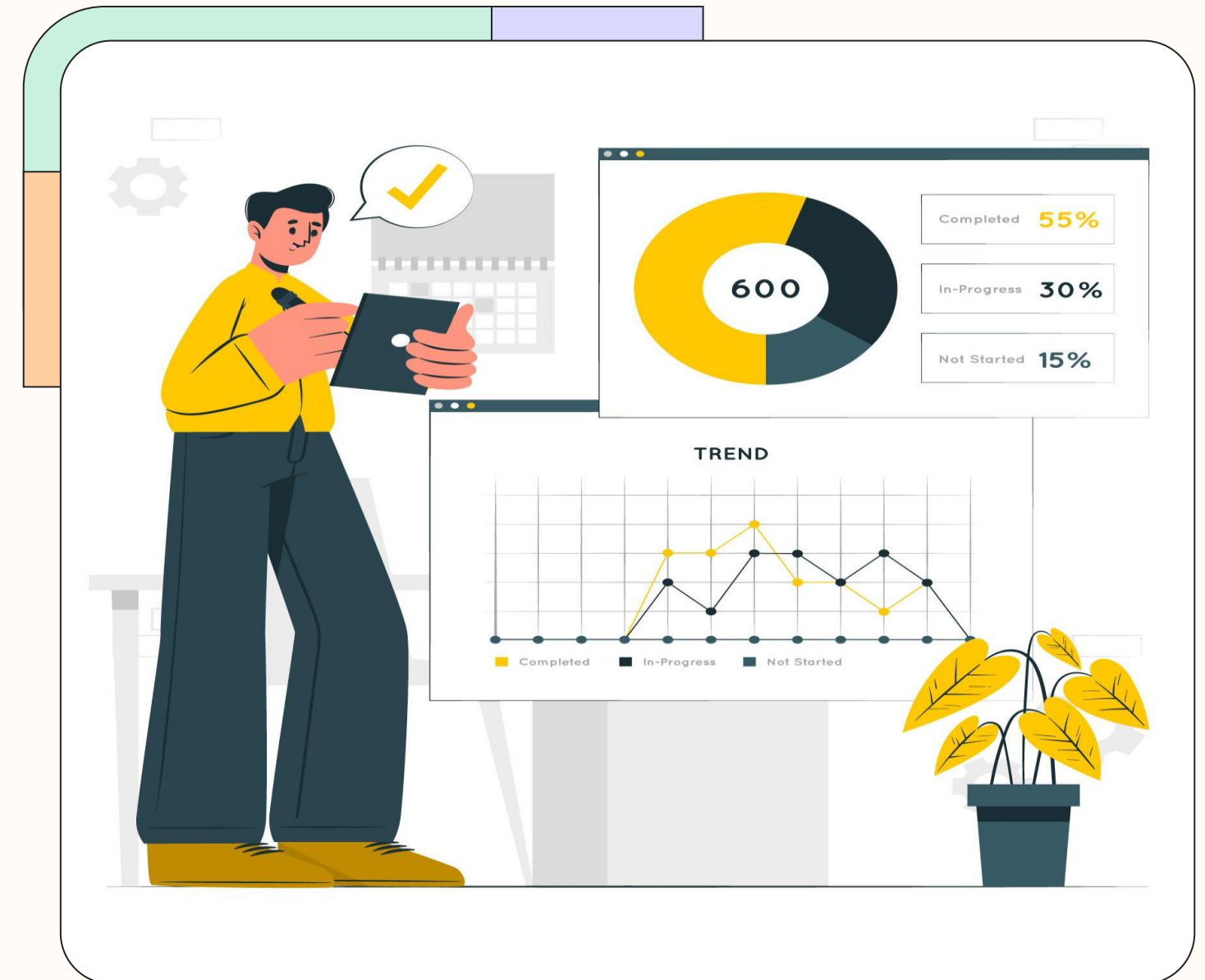


Introduction

DigiBill is a modern Android app designed for shopkeepers to manage billing and inventory digitally. It replaces manual invoicing with fast, secure, and paperless transactions using features like barcode scanning, real-time stock updates, and cloud-based data storage. With DigiBill, businesses can create and share invoices instantly via SMS, track sales, and manage inventory—all from their smartphone. It offers a simple, efficient, and eco-friendly solution for everyday billing needs.

Project Overview

DigiBill is an Android-based mobile application designed to simplify and modernize invoice management for shopkeepers. The app enables paperless, fast, and secure billing by integrating features like barcode scanning, real-time inventory tracking, and automated GST calculations. Built using Kotlin(Android Studio) and powered by Firebase Firestore, DigiBill ensures real-time data synchronization, secure cloud storage, and instant invoice sharing via SMS.



PROJECT MANAGEMENT ROADMAP



Project Planning

Define the goals and project schedule before starting.



Execution

Execute the project according to the plan and monitor progress.



Completion

Complete all deliverables and conduct project evaluation.



Team Allocation

Allocate resources and the team that will work on the project.



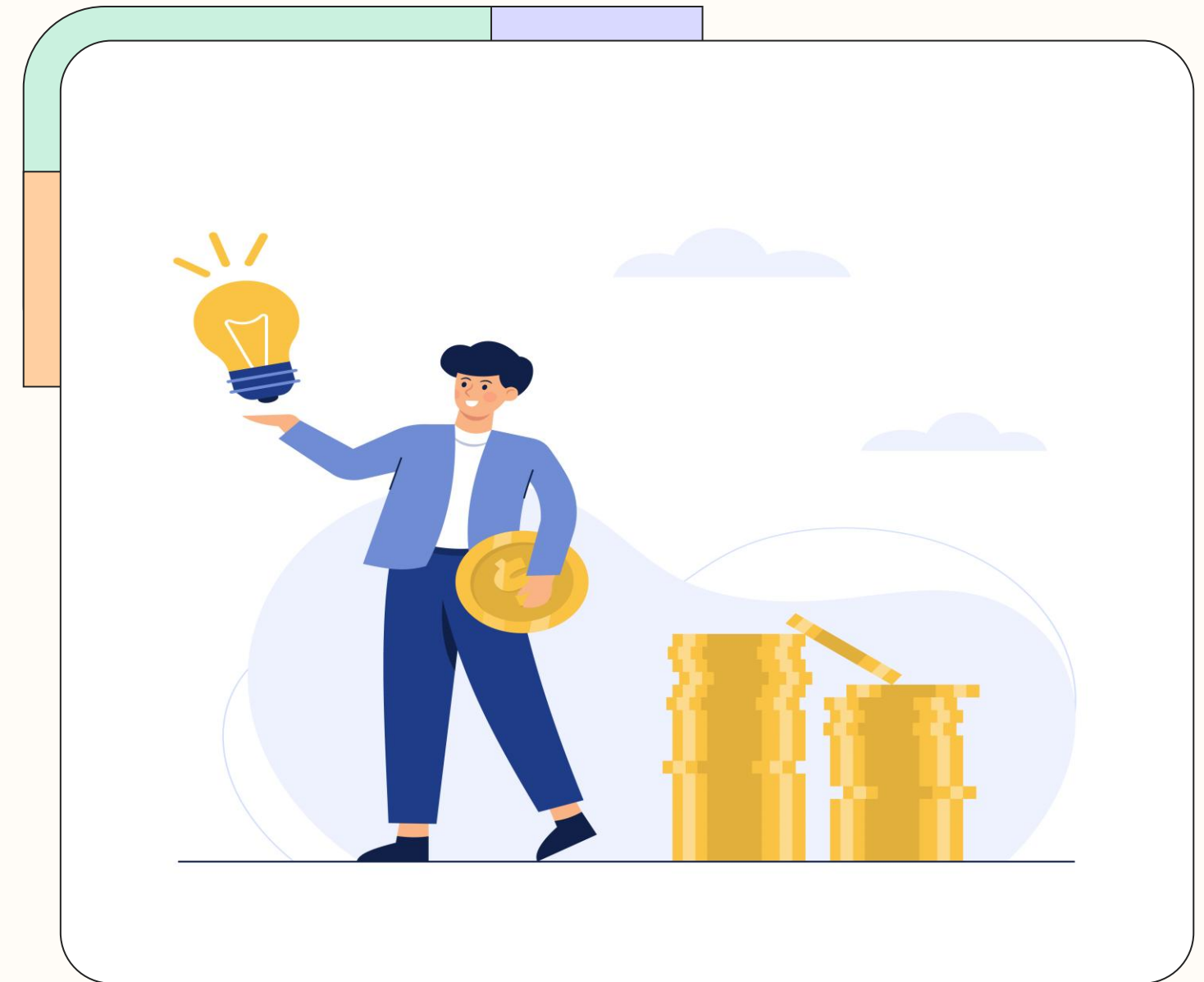
Monitoring

Regularly track project progress to ensure goals are met.



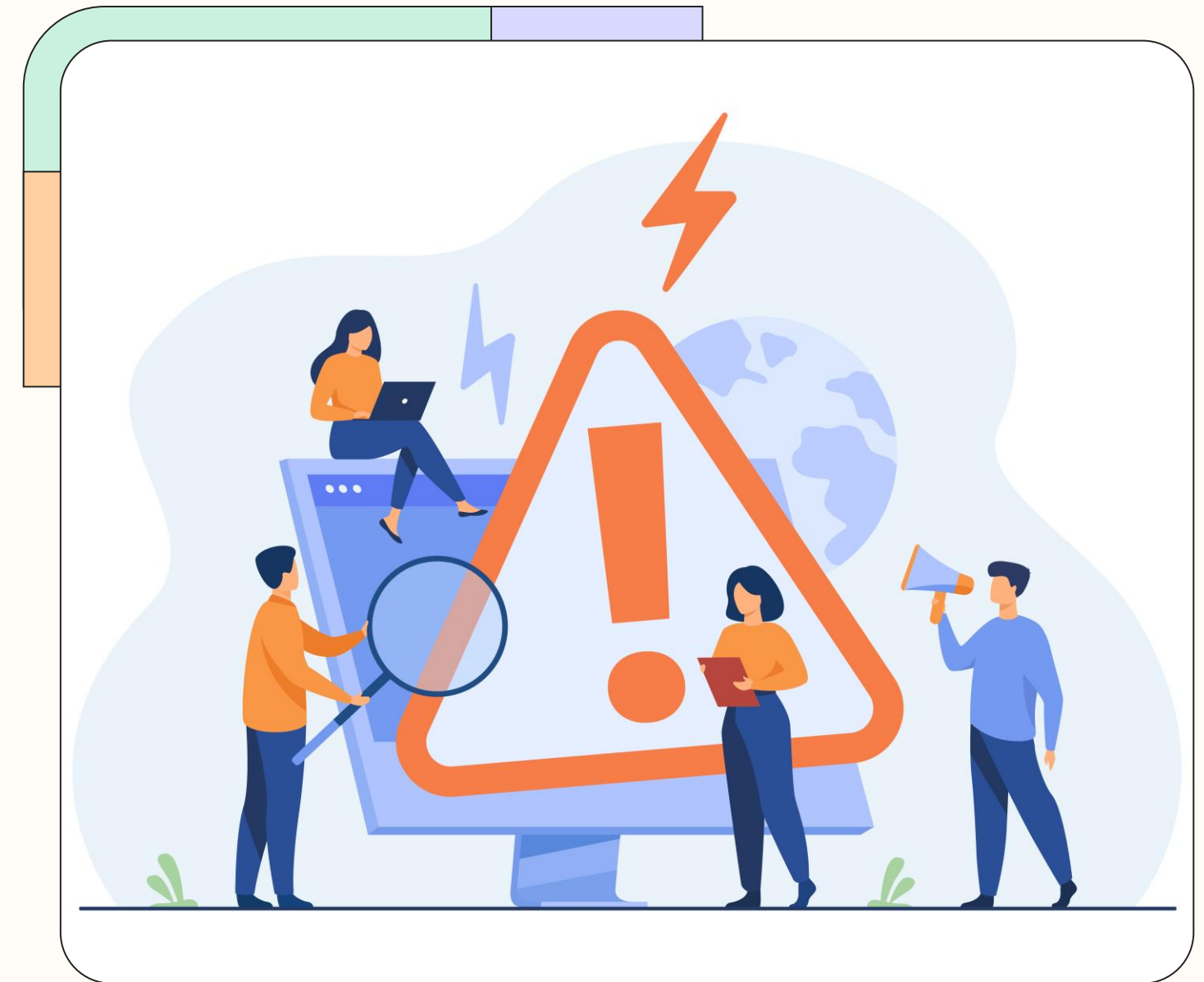
Benefits

- **Fast Billing:** Quick invoice creation with barcode scanning and automated GST.
- **Real-Time Inventory:** Stock updates instantly after each sale to avoid shortages.
- **Cloud Storage:** Data securely stored and synced via Firebase, accessible anytime.
- **Paperless Invoicing:** Share invoices digitally via SMS, reducing paper use.
- **Easy to Use:** Simple UI designed for shopkeepers, works on most Android devices.



Limitations

- **Android Only:** Not available on iOS or web platforms.
- **No Payment Gateway:** Doesn't support online payments (UPI, cards, etc.).
- **Internet Required:** Needs internet for syncing; offline mode is limited.
- **No Email Sharing:** Invoices can't be sent via email—only SMS.
- **Device Requirements:** Requires Android 7+ with at least 4GB RAM and a working camera.



Activity Diagram

The **Activity Diagram** page provides a comprehensive visualization of how users move through the system's interface and interact with various components. It serves as a roadmap for understanding the overall user experience, ensuring intuitive and efficient navigation across different sections of the application. By mapping out the pathways between screens, actions, and decision points, this diagram helps in identifying usability issues, streamlining workflows, and enhancing accessibility. Additionally, it plays a crucial role in optimizing the interface structure, improving user engagement, and guiding developers and designers in creating a seamless and user-friendly application experience.



Activity Diagram



Get detailed view here.

Use Case Diagram

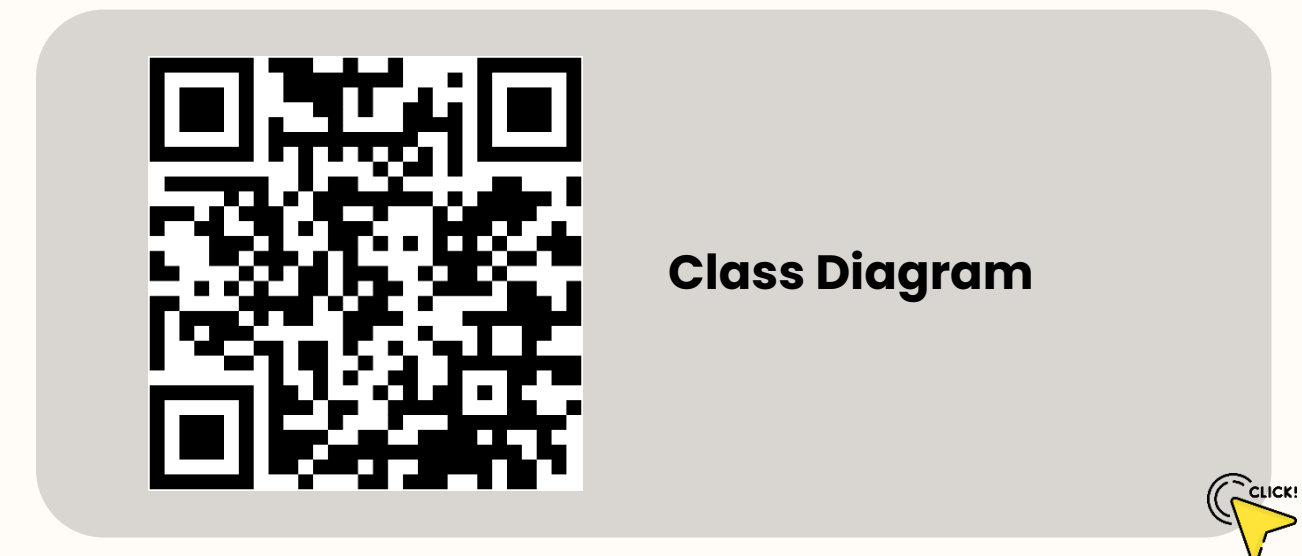
The **Use Case Diagram** page provides a structured visual representation of the interactions between various system components and external actors. It serves as a foundational blueprint for understanding the system's functional requirements, depicting how users and external entities engage with different functionalities. By illustrating key use cases, user roles, and system boundaries, this diagram helps in defining user workflows, identifying essential features, and ensuring that all functional requirements are addressed. Additionally, it aids in system analysis, requirement validation, and effective communication among stakeholders, contributing to a well-structured and user-centric system design.



Get detailed view here.

Class Diagram

The **Class Diagram** page visually represents the movement of data within the system, illustrating how information flows between various processes, data stores, and external entities. It serves as a crucial tool for understanding data processing, system functionality, and the interaction between different components. By providing a clear and structured overview of data inputs, transformations, and outputs, the DFD helps in identifying inefficiencies, optimizing workflows, and ensuring smooth data exchange. Additionally, it aids in system analysis, documentation, and communication among developers, designers, and stakeholders, contributing to a well-structured and efficient system architecture.



Get detailed view here.

Data dictionary

The **Data Dictionary** page provides a structured and detailed representation of the system's data entities, their attributes, and the relationships between them. It serves as a foundational blueprint for understanding how different entities interact, how data is organized, and how relationships are established within the system. This diagram plays a crucial role in defining the system's database structure, ensuring data integrity, consistency, and efficient management. Additionally, it aids in visualizing complex data relationships, optimizing database design, and facilitating smooth data retrieval and manipulation. By clearly mapping out the system's data flow, the ER diagram enhances scalability, maintainability, and overall system performance.



Get detailed view here.

User Interface

The **User Interface (UI) Design Document** provides a structured representation of the system's visual elements, user interactions, and navigation flow. It serves as a blueprint for understanding how different interface components are designed, how users interact with the system, and how various UI elements are structured to enhance usability. This document helps in defining the system's UI layout, ensuring a seamless, intuitive, and user-friendly experience. Additionally, it aids in maintaining consistency across different screens, improving accessibility, and optimizing user workflows for better engagement and efficiency.



Get detailed view here.

Application Testing

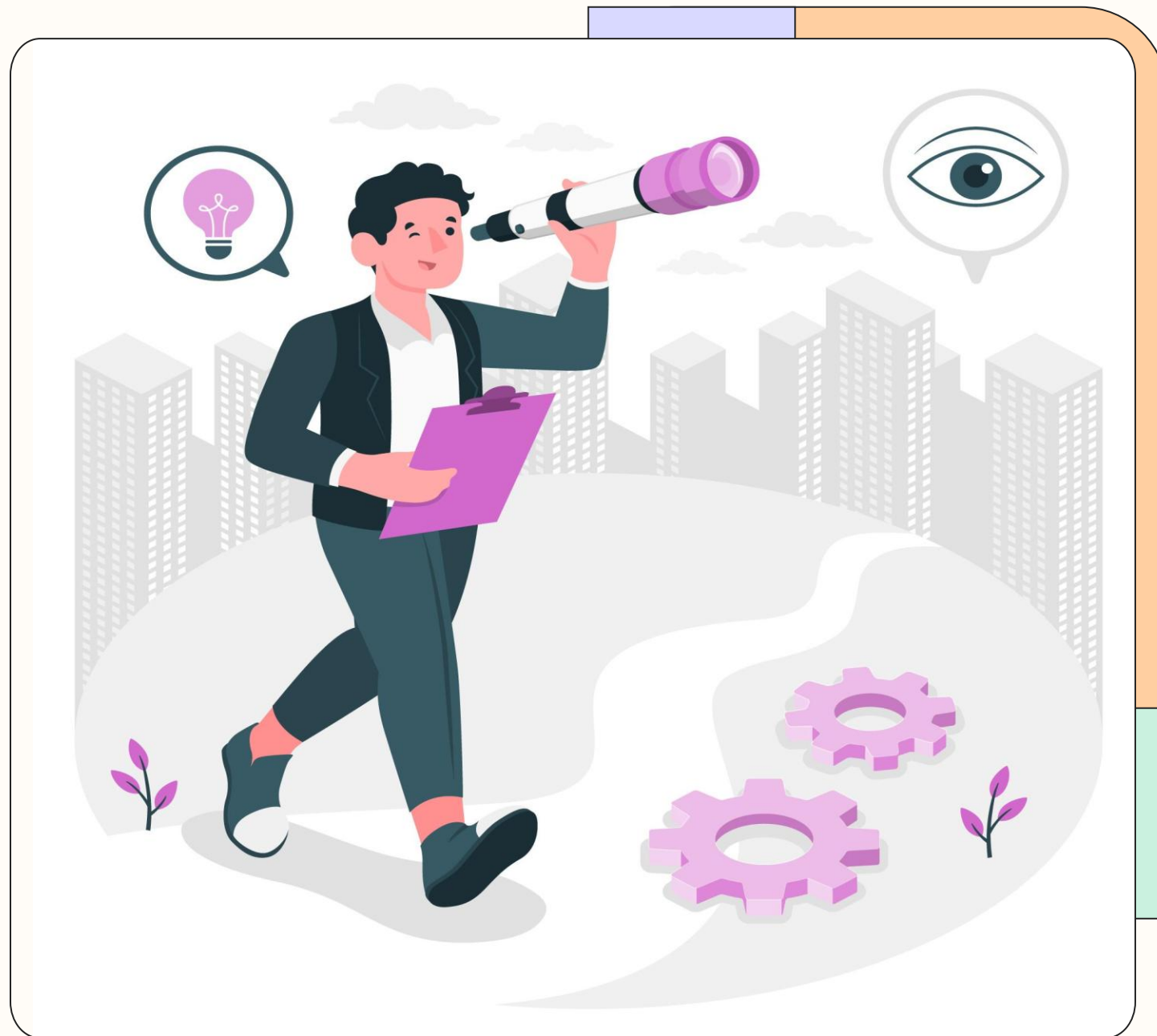
Application Testing ensures that **DigiBill** meets its business requirements and performs its intended operations accurately and efficiently. It plays a crucial role in identifying and resolving defects, ensuring system reliability, and enhancing user experience. The testing process involves multiple layers of evaluation, each focusing on specific aspects of the application's functionality, security, and performance.

Types of Testing:

- **Unit Testing** – Verifies individual components or modules of the application to ensure they function correctly in isolation.
- **Functional Testing** – Examines whether the system behaves as expected by validating input, output, and user interactions.
- **Security Testing** – Assesses vulnerabilities to safeguard the application from threats, unauthorized access, and data breaches.



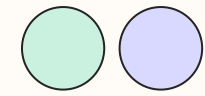
Get detailed view here.



Future Scope & Planning

DigiBill aims to enhance accessibility, efficiency, and user convenience with offline support, cross-platform availability, integrated payments, and automated SMS notifications, ensuring a seamless digital billing experience.

- Offline Support: Smooth offline operations with auto-sync.
- Platform Independence: Expand to iOS and web.
- Payment Gateway: Enable UPI, cards, and wallet payments.
- SMS Integration: Automate invoices and notifications.



Conclusion and Q&A

DigiBill is not just an invoicing tool, it is a complete digital billing solution that helps businesses save time, reduce costs, enhance customer experience, and transition to a fully paperless environment. With its robust architecture, seamless functionality, and future scalability, DigiBill is set to revolutionize the way shopkeepers manage their sales and inventory in the digital era.



Thank You